

Cross-Benchmark Analysis of Cultural Commonsense Knowledge in Language Models

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Abstract

As large language models (LLMs) are deployed globally, evaluating their ability to represent diverse cultures has become critical. Due to their predominantly Western-centric training data, current LLMs risk systematically underrepresenting or marginalizing many global regions and cultures. This has motivated the development of cultural knowledge benchmarks designed to assess model performance across different cultural contexts. However, a fundamental question remains underexplored: do these benchmarks measure the same underlying constructs, or do they capture distinct aspects of cultural knowledge? We address this gap by analyzing correlations across four diverse cultural knowledge evaluation frameworks: DLAMA-v1, CulturalBench, BLEnD, and GeoMLAMA, focusing specifically on common sense knowledge. We evaluate two 3B-parameter multilingual models (Llama 3.2-3B and Qwen 2.5-3B) and conduct correlation analyses at both the benchmark level and country level across common countries. Our findings reveal weak to moderate correlations (Pearson $r < 0.4$) between benchmarks, indicating they capture distinct facets of cultural knowledge rather than a unified construct. These results challenge the assumption that cultural benchmarks are interchangeable and underscore the necessity of multi-benchmark evaluation for a comprehensive assessment of cultural knowledge in language models.¹

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across diverse tasks, yet their performance on culturally grounded knowledge remains uneven. As these models are increasingly deployed in global applications, from education and customer service to content generation and decision support, their ability to ac-

curately represent diverse cultures carries significant real-world consequences. Recent evaluations have demonstrated that widely-used LLMs exhibit systematic cultural bias, with outputs most closely aligned with English-speaking and European countries while showing substantial distance from African-Islamic and other non-Western cultural values (Tao et al., 2024). Research has shown that AI usage patterns and performance can vary significantly across geocultures and demographic groups (Shelby et al., 2025), with culturally biased models potentially failing systematically for certain user populations.

In response, multiple benchmarks have been developed to evaluate cultural knowledge in LLMs. However, a fundamental question remains underexplored: do these benchmarks measure the same underlying constructs, or do they assess distinct facets of cultural understanding? Prior work suggests that cultural knowledge is inherently multidimensional, and that benchmarks tend to isolate only one or two cultural facets at a time (Adilazuarda et al., 2024). In this paper, we focus specifically on cultural benchmarks that evaluate factual or commonsense cultural knowledge, such as foods, customs, and geographically grounded cultural practices. This distinction is important, as these benchmarks differ from those measuring cultural values, norms, or stereotypes.

Given this distinction, we need to understand whether these commonsense cultural benchmarks align or whether each one measures something different. If benchmarks designed to test similar knowledge disagree, this suggests they define cultural knowledge in different ways.

This question has important implications for both model evaluation and development. If benchmarks are highly correlated, researchers can confidently select a single benchmark for evaluation. Conversely, low correlations suggest that comprehensive cultural assessment requires multiple com-

¹Code and evaluation pipeline available at: <https://github.com/ameme12/vllm-demo>

plementary benchmarks, each capturing different dimensions of cultural knowledge. Without understanding benchmark construct validity, we risk drawing incomplete conclusions about model cultural competence and may fail to accurately identify quality-of-service, allocative, or representational harms that disproportionately affect certain user populations (Shelby et al., 2025).

We address this gap through systematic cross-benchmark correlation analysis, examining four prominent cultural evaluation frameworks: BLEnD (Myung et al., 2025), CulturalBench (Chiu et al., 2024), GeoMLAMA (Yin et al., 2022), and DLAMA-v1 (Keleg and Magdy, 2023). We evaluate two 3B-parameter multilingual models and analyze correlations both at the benchmark level and at the country level where overlapping coverage exists.

2 Related Work

The growing deployment of large language models globally has motivated extensive research on cultural awareness in NLP systems. Pawar et al. (Pawar et al., 2024) provide the most comprehensive survey to date, examining over 300 papers across text, vision, audio, and video modalities. They catalog benchmarks across eight domains including academic knowledge (MMLU-series), commonsense knowledge (monocultural: COPAL-ID, IndoCulture, CLiCK, HAE-RAE; multicultural: BLEnD, GEOMLAMA, CULTURE-GEN, CUNIT, CultureAtlas, DLAMA-v1), social values (World-ValuesBench, CDEval), social norms and morals (NORMBANK, NORMDIAL, CultureBank, EtiCor), and social biases (BBQ-series, CrowS-Pairs variants). While their survey systematically compares data creation methodologies and alignment techniques across these benchmarks, it stops short of empirically investigating whether benchmarks claiming to evaluate the same cultural construct actually converge in their assessments.

AlKhamissi et al. (AlKhamissi et al., 2024) complement this taxonomic work through qualitative analysis of 20 datasets, proposing a four-part framework distinguishing culture-as-knowledge (factual recall), culture-as-preference (survey-derived values), culture-as-dynamics (context-dependent enactment), and culture-as-bias (stereotype-based harm). While their taxonomy reveals that different benchmarks conceptualize culture through divergent theoretical lenses (e.g., BLEND’s culture-as-

knowledge versus CASA’s culture-as-preference), they do not empirically test whether benchmarks said to measure the same dimension actually converge in their assessments.

A fundamental question remains empirically unexplored: do cultural benchmarks designed to measure similar constructs actually produce correlated results?

We address this gap through systematic cross-benchmark correlation analysis, examining four prominent cultural commonsense knowledge evaluation frameworks selected from Pawar et al. (Pawar et al., 2024) survey, which were identified as testing similar aspects of culture: BLEnD (Myung et al., 2025), CulturalBench (Chiu et al., 2024), GeoMLAMA (Yin et al., 2022), and DLAMA-v1 (Keleg and Magdy, 2023). By evaluating two 3B-parameter multilingual models (Llama 3.2-3B and Qwen 2.5-3B) and analyzing correlations at both benchmark and country levels, we provide the first empirical evidence of how cultural knowledge benchmarks converge, or fail to converge, in their assessments. Our findings reveal weak to moderate correlations (Pearson $r < 0.4$), suggesting that these benchmarks capture distinct facets of cultural knowledge rather than a unified construct, with important implications for how the community should approach comprehensive cultural evaluation in practice.

3 Datasets and Evaluation Metrics

3.1 Benchmark Selection Methodology

We base our benchmark selection on the comprehensive taxonomy presented in Pawar et al. (2024), which surveys over 300 papers and benchmarks on cultural awareness in language models. Following their classification framework (Figure 5), we focus specifically on the *commonsense knowledge* category, which evaluates models’ understanding of everyday cultural facts, practices, and region-specific knowledge rather than cultural values or biases.

Within this category, we select benchmarks that satisfy three key criteria: (1) *multicultural coverage* - evaluation across multiple world regions to enable cross-benchmark comparison, (2) *explicit geographic grounding* - clear association of questions with specific countries or regions, and (3) *established evaluation protocols* - well-documented methodology and publicly available datasets. This selection strategy yields four diverse benchmarks

that span different evaluation formats (multiple-choice, cloze-completion, and generation), geographic scopes (4-45 countries), and cultural knowledge types (everyday practices, factual knowledge, and commonsense reasoning).

3.2 Datasets

We select four cultural evaluation benchmarks that represent diverse approaches to assessing cultural commonsense knowledge in LLMs:

BLEnD (Myung et al., 2025): A hand-crafted benchmark of 52.6k question-answer pairs spanning 16 countries/regions, targeting everyday cultural knowledge including foods, activities, customs, and lived practices. Questions were authored by native speakers or cultural insiders. We evaluate the multiple-choice format.

CulturalBench (Chiu et al., 2024): Consists of 1,227 human-written questions assessing cultural knowledge across 45 global regions. We evaluate CulturalBench-Easy, which uses a multiple-choice format with four options.

GeoMLAMA (Yin et al., 2022): A multilingual cloze-style benchmark evaluating geo-specific commonsense knowledge. The dataset includes fill-in-the-blank prompts covering everyday cultural facts such as foods, holidays, currency, and social practices for specific countries. Available in five languages, we use the English subset reformatted as multiple-choice.

DLAMA-v1 (Keleg and Magdy, 2023): A culturally balanced factual knowledge benchmark with 78,259 Wikidata triples across three culturally contrasting region pairs: Arab-Western, Asian-Western, and South American-Western. Unlike the other benchmarks, DLAMA-v1 focuses on factual triples (e.g., "The capital of X is Y") and uses generation-based evaluation.

3.3 Evaluation Metrics

We employ accuracy as the primary evaluation metric, enabling direct cross-benchmark comparison. We assess construct validity through Pearson correlation analysis from two complementary perspectives: (1) *benchmark-level correlation* examines whether benchmarks produce consistent country rankings by correlating country-level accuracy scores across benchmark pairs, and (2) *country-level correlation* examines whether country pairs

show similar performance patterns across benchmarks by correlating benchmark-level accuracy scores. Both analyses use only countries with overlapping geographic coverage.

4 Experimental Setup

4.1 Models

We evaluate two multilingual instruction-tuned language models in the 3B parameter range. Due to computational constraints, larger architectures were not feasible on our available hardware. Despite their smaller size, these models enable examination of how multilingual training influences geo-diverse commonsense knowledge.

Llama 3.2-3B Instruct (Meta AI, 2024): This 3B-parameter model is trained on a mixture of English and non-English text, offering broad multilingual exposure. Its training data diversity makes it suitable for evaluating how smaller LLMs acquire culturally grounded commonsense across languages.

Qwen2.5-3B Instruct (Team, 2024): A 3B-parameter multilingual model designed for cross-lingual understanding and culturally diverse reasoning. Its architecture and training emphasize robust multilingual performance, providing a complementary perspective on cultural knowledge acquisition in low-parameter LLMs.

4.2 Evaluation Pipeline

We developed a modular Python-based evaluation framework implementing a standardized BaseTask interface for each benchmark. The pipeline handles data loading, prompt formatting, model inference via vLLM, and metric computation while preserving each benchmark’s original conventions. All configurations are specified in YAML config files for reproducibility.

4.3 Task Formatting and Prompting

To enable direct cross-benchmark comparison, we standardize BLEnD, CulturalBench, and GeoMLAMA as multiple-choice questions with four options (A-D), requesting JSON-formatted responses: `{"answer_choice":""}`. DLAMA-v1 uses its original generation-based format with LLM-as-judge evaluation.

BLEnD (Myung et al., 2025): We evaluate the *multiple-choice-questions* subset from HuggingFace (306,370 test samples across 16 cultures) at

the country level.

CulturalBench (Chiu et al., 2024): We evaluate *CulturalBench-Easy* (1,230 test samples across 45 countries) at the region level as specified by the benchmark, computing country-level scores where sufficient data exists for cross-benchmark correlation analysis.

GeoMLAMA (Yin et al., 2022): Originally a cloze-completion task, we reformat GeoMLAMA into multiple-choice format using the benchmark’s provided *candidate_answers*. We retrieve the dataset from the original GitHub repository², using the English subset (150 base prompts) augmented following the benchmark’s methodology.

Example JSON-based prompt:

What is a common snack for preschool kids in Algeria? Without any explanation, choose only one from the given alphabet choices (e.g., A, B, C). Provide as JSON format: {"answer_choice":""}

- A. chocolate paste
- B. egg
- C. fruit
- D. jam sandwiches

Answer:

DLAMA-v1 (Keleg and Magdy, 2023): We evaluate the English split (*Asian-West* subset but due to time constraints were unable to evaluate *Arab-West*) using the original generation-based protocol with LLM-as-judge evaluation (allenai/OLMo-2-1124-7B-Instruct). The judge assigns binary scores based on semantic equivalence to acceptable answers. Performance is computed at the region level, with country-level accuracy calculated using Wikipedia country identifiers for cross-benchmark correlation analysis.

All benchmarks are evaluated in English to isolate cultural knowledge assessment from language proficiency effects.

4.4 Response Parsing and Evaluation

Multiple-Choice Benchmarks (BLENd, CulturalBench, GeoMLAMA): We extract answer choices using a three-stage pipeline: (1) JSON format parsing, (2) standalone letter detection, (3) fallback pattern matching. Responses receive binary scores (1 for correct, 0 for incorrect). For GeoMLAMA, we additionally map extracted letters to

answer text and verify against acceptable answer lists using case-insensitive string matching.

DLAMA-v1 Generation Evaluation: DLAMA-v1 uses free-form generation evaluated via LLM-as-judge (allenai/OLMo-2-1124-7B-Instruct). The judge determines whether generated responses semantically match acceptable answers from the benchmark’s gold list, assigning binary scores (1 for TRUE, 0 for FALSE). The evaluation accepts dialectal variations (e.g., "Moroccan Arabic" = "Arabic") and common synonyms (e.g., "USA" = "United States") while ignoring hedging language. We acknowledge that judge model errors may introduce noise compared to the rule-based evaluation used for other benchmarks.

We report accuracy as the primary evaluation metric for all benchmarks, computed as the mean score across all samples. We calculate accuracy at both the overall benchmark level and at the country level to enable our two-tier correlation analysis approach described in Section 4.6.

4.5 Implementation Details

Infrastructure: We use vLLM (v0.6.1.post2) for batched inference with temperature=0.0 (greedy decoding), batch size 32 (16 for DLAMA), and max 100 tokens for MCQ tasks. Experiments ran on Mila cluster NVIDIA L40S GPUs with 64-256GB RAM. Results are saved as JSON files with per-sample predictions and aggregate metrics. Random seed fixed at 42 for reproducibility.

Complete experimental configurations and code are available at <https://github.com/ameme12/vllm-demo>.

4.6 Cross-Benchmark Correlation Analysis

To assess whether cultural benchmarks measure the same underlying construct, we compute pairwise Pearson correlations at the country level. Due to limited geographic overlap, we conduct two three-way comparisons: (1) BLENd, CulturalBench, and GeoMLAMA across 4 countries (US, China, India, Iran), and (2) BLENd, CulturalBench, and DLAMA-v1 across 6 countries (China, Indonesia, South Korea, Spain, UK, US).

For each comparison, we compute two correlation matrices:

Benchmark Correlation: Correlates country-level accuracy scores across benchmark pairs to assess

²<https://github.com/WadeYin9712/GeoMLAMA>

whether benchmarks produce consistent country rankings (n = number of common countries).

Country Correlation: Correlates benchmark-level accuracy scores across country pairs to assess whether countries show similar performance patterns (n = 3 benchmarks for both comparisons).

Both analyses are conducted separately for each model. We exclude countries with <10 samples per benchmark. Results visualized as correlation heatmaps with diverging colormaps.

Interpretation: Strong correlations ($r > 0.7$) suggest benchmarks measure similar constructs; moderate ($0.4 < r < 0.7$) suggest overlapping but distinct aspects; weak ($r < 0.4$) indicate distinct measurement dimensions requiring multi-benchmark evaluation.

5 Results and Discussion

We conduct two three-way benchmark comparisons due to limited geographic overlap: (1) BLEnD, CulturalBench, and GeoMLAMA across 4 common countries (US, China, India, Iran), and (2) BLEnD, CulturalBench, and DLAMA-v1 across 6 common countries (US, UK, China, Spain, Indonesia, South Korea).

5.1 Benchmark-Level Correlation Analysis

Benchmark-level correlations assess whether different cultural evaluation frameworks produce consistent country rankings. Strong correlations ($r > 0.7$) would indicate that benchmarks measure similar aspects of cultural knowledge and may be interchangeable, while weak correlations ($r < 0.4$) would suggest that benchmarks capture distinct facets and comprehensive evaluation requires multiple benchmarks.

5.1.1 BLEnD, CulturalBench, and GeoMLAMA

Figures 1 and 2 present benchmark correlation matrices for both models across four common countries (US, China, India, Iran). The results show that benchmark agreement varies substantially between models.

For **Qwen 2.5-3B**, we observe strong correlations between CulturalBench and both other benchmarks ($r = 0.834$ with BLEnD, $r = 0.850$ with GeoMLAMA), suggesting these benchmarks produce highly consistent country rankings for this model. However, the BLEnD-GeoMLAMA correlation is

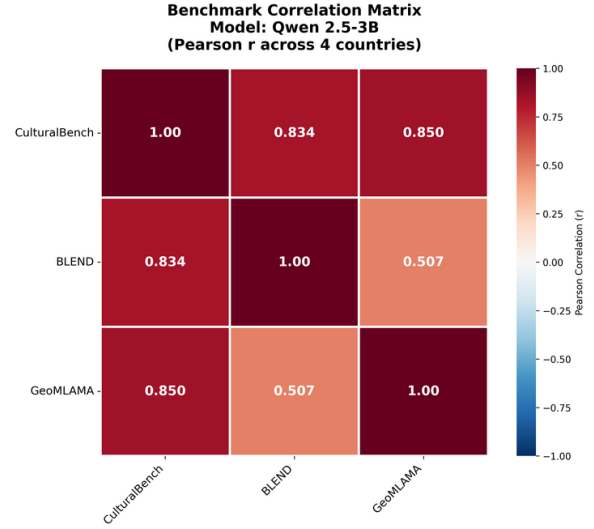


Figure 1: Benchmark-level correlation matrix for Qwen 2.5-3B across BLEnD, CulturalBench, and GeoMLAMA. Each cell shows the Pearson correlation coefficient between benchmark pairs based on their country-level accuracy scores across 4 common countries (US, China, India, Iran).

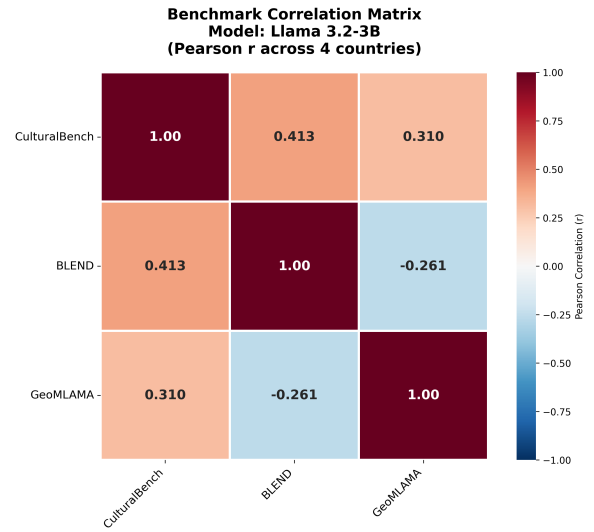


Figure 2: Benchmark-level correlation matrix for Llama 3.2-3B across BLEnD, CulturalBench, and GeoMLAMA. Note the negative correlation ($r = -0.261$) between BLEnD and GeoMLAMA, indicating these benchmarks produce inverse country rankings for this model.

notably weaker ($r = 0.507$), falling in the moderate range and indicating these benchmarks may assess different aspects of cultural knowledge for the same four countries.

In contrast, **Llama 3.2-3B** exhibits substantially weaker benchmark agreement overall. The CulturalBench-BLEnD correlation drops

to $r = 0.413$ (moderate), while CulturalBench-GeoMLAMA falls to $r = 0.310$ (weak). Most strikingly, BLEnD and GeoMLAMA show a weak *negative* correlation ($r = -0.261$), meaning countries that perform well on BLEnD for Llama tend to perform poorly on GeoMLAMA, and vice versa.

To illustrate this inverse relationship, Figure 15 presents a detailed performance comparison showing how the same countries rank differently across these two benchmarks. The United States ranks 1st on BLEnD (87.8%) but drops substantially on GeoMLAMA (44.8%), while China shows the opposite pattern, ranking 3rd on BLEnD (79.6%) but 1st on GeoMLAMA (55.2%). This inverse relationship suggests these benchmarks either (1) capture fundamentally different dimensions of cultural knowledge (everyday practices vs. geographically-grounded facts), or (2) trigger opposing biases in Llama’s training data that cause the model to excel on one benchmark’s format while failing on the other’s.

5.1.2 BLEnD, CulturalBench, and DLAMA-v1

Figures 3 and 4 present the benchmark correlation matrices for both models across six common countries (US, UK, China, Spain, Indonesia, South Korea). Despite the larger sample size compared to the GeoMLAMA comparison ($n=6$ vs. $n=4$), the results reveal similarly weak correlations, with notable divergence between models.

For **Qwen 2.5-3B**, we observe a very strong correlation between CulturalBench and BLEnD ($r = 0.895^*$), suggesting these benchmarks produce highly consistent country rankings for this model across the six evaluated countries. However, DLAMA-v1 shows weak correlations with both benchmarks: $r = 0.264$ with CulturalBench and $r = -0.144$ with BLEnD. The weak negative BLEnD-DLAMA correlation, suggests these benchmarks may rank countries in somewhat opposing orders.

Llama 3.2-3B exhibits a similar pattern but with an even stronger CulturalBench-BLEnD correlation ($r = 0.899^*$, $p < 0.05$), indicating near-perfect agreement between these two benchmarks for this model. However, DLAMA-v1 again diverges substantially: the CulturalBench-DLAMA correlation is weak ($r = 0.166$), and the BLEnD-DLAMA correlation is weakly negative ($r = -0.162$). This pattern mirrors the GeoMLAMA results, where DLAMA-v1 appears to measure a distinct dimension of cultural knowledge that is largely independent to what

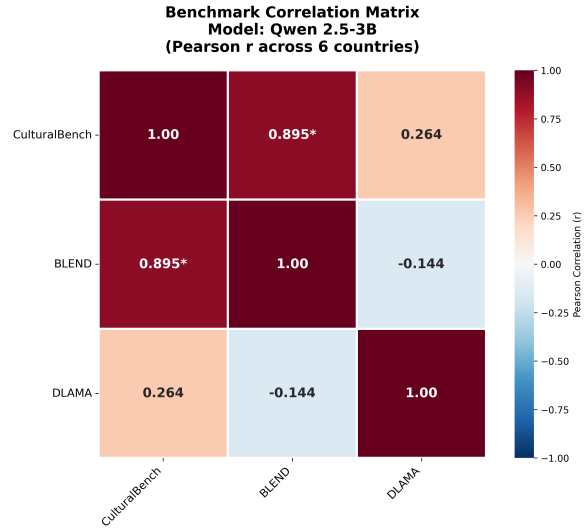


Figure 3: Benchmark-level correlation matrix for Qwen 2.5-3B across BLEnD, CulturalBench, and DLAMA-v1. Each cell shows the Pearson correlation coefficient between benchmark pairs based on their country-level accuracy scores across 6 common countries (US, UK, China, Spain, Indonesia, South Korea).

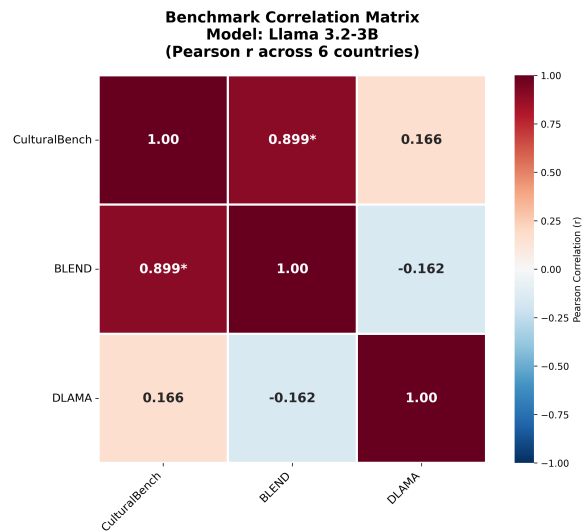


Figure 4: Benchmark-level correlation matrix for Llama 3.2-3B across BLEnD, CulturalBench, and DLAMA-v1. Note the weak negative correlations between DLAMA-v1 and both other benchmarks.

BLEnD and CulturalBench assess.

One potential explanation for DLAMA-v1’s divergence is its distinct evaluation approach. Unlike BLEnD and CulturalBench, which use multiple-choice formats with explicit answer options, DLAMA-v1 employs free-form generation evaluated via LLM-as-judge. This methodological difference may privilege different types of knowledge retrieval: multiple-choice formats may favor

recognition-based knowledge (identifying the correct answer among alternatives), while generation formats require unprompted recall. Additionally, DLAMA-v1 focuses specifically on factual Wiki-data triples (e.g., "The capital of X is Y"), whereas BLEnD and CulturalBench include more experiential and practice-based questions (e.g., common foods, daily customs). This fundamental difference in what constitutes "cultural knowledge" likely contributes to the weak correlations.

Despite having more countries in this comparison ($n=6$ vs. $n=4$ for GeoMLAMA), only the CulturalBench-BLEnD correlation reaches statistical significance in each matrix. This reflects a fundamental limitation of our analysis: although these benchmarks collectively evaluate 45+ countries, only a small number have overlapping coverage, restricting our ability to draw statistically robust conclusions about benchmark relationships.

5.2 Country-Level Correlation Analysis

While benchmark-level correlations reveal how consistently benchmarks rank countries, country-level correlations examine whether country pairs exhibit similar performance patterns across benchmarks. High correlations (e.g., China-Iran: $r = 0.99$) indicate countries experience similar relative performance across benchmarks, while low correlations suggest independent patterns that may reflect different evaluation approaches for different cultural contexts.

5.2.1 Performance Patterns Across BLEnD, CulturalBench, and GeoMLAMA

Figures 5 and 6 present country correlation matrices for both models across the three benchmarks. These matrices reveal high within-country consistency for some country pairs but substantial variation for others, with notable model-dependent patterns.

Llama 3.2-3B exhibits the following patterns:

- **China-Iran:** Near-perfect correlation ($r = 0.99$), indicating these countries show nearly identical relative performance patterns across all three benchmarks
- **India-United States:** Strong correlation ($r = 0.94$), suggesting similar cross-benchmark performance profiles
- **Weaker cross-regional correlations:** China-India ($r = 0.32$) and India-Iran ($r = 0.47$) show

much weaker relationships, indicating these country pairs have divergent performance patterns across benchmarks

Qwen 2.5-3B exhibits similar but intensified patterns:

- **China-Iran:** Perfect correlation ($r = 1.00$), meaning Qwen's performance rankings for these two countries are identical across all three benchmarks
- **India-Iran:** Very strong correlation ($r = 0.82$)
- **China-India:** Strong correlation ($r = 0.77$)
- **India-United States:** Zero correlation ($r = 0.00$), indicating completely independent performance patterns across benchmarks for this country pair

The high correlations between China and Iran ($r = 0.99$ - 1.00 across both models) reveal that these countries exhibit nearly identical performance profiles across benchmarks. Examining Figure 15, both countries show consistent patterns: relatively strong performance on BLEND (China: 79.6%, Iran: 74.4% as displayed in Figure 9) but substantial declines on GeoMLAMA (China: 55.2%, Iran: 36.0% as displayed in Figure 11), with Iran's drop being particularly pronounced. Both countries also show intermediate performance on CulturalBench (China: 57.6%, Iran: 46.0% as displayed in Table 2). This parallel pattern, which is strong on everyday cultural practices (BLEND), weaker on geographically-grounded facts (GeoMLAMA), suggests that the benchmarks assess Chinese and Iranian cultural knowledge through similar evaluative lenses. Whatever makes GeoMLAMA challenging for China also makes it challenging for Iran, and vice versa.

This consistency could reflect several factors: (1) similar underrepresentation in training data, with both being non-Western, non-English-dominant regions; (2) similar cultural distance from Western-centric knowledge bases embedded in model parameters; or (3) similar types or structures of cultural knowledge being tested across benchmarks for these regions (e.g., both may be evaluated through similar question formats or knowledge domains).

In stark contrast, other country pairs show divergent and model-dependent patterns. Most strikingly, while Llama exhibits strong India-US correlation ($r = 0.94$), indicating similar performance

Country Correlation Matrix
Model: Qwen 2.5-3B
(CulturalBench, BLEND, GeoMLAMA, 3 benchmarks)

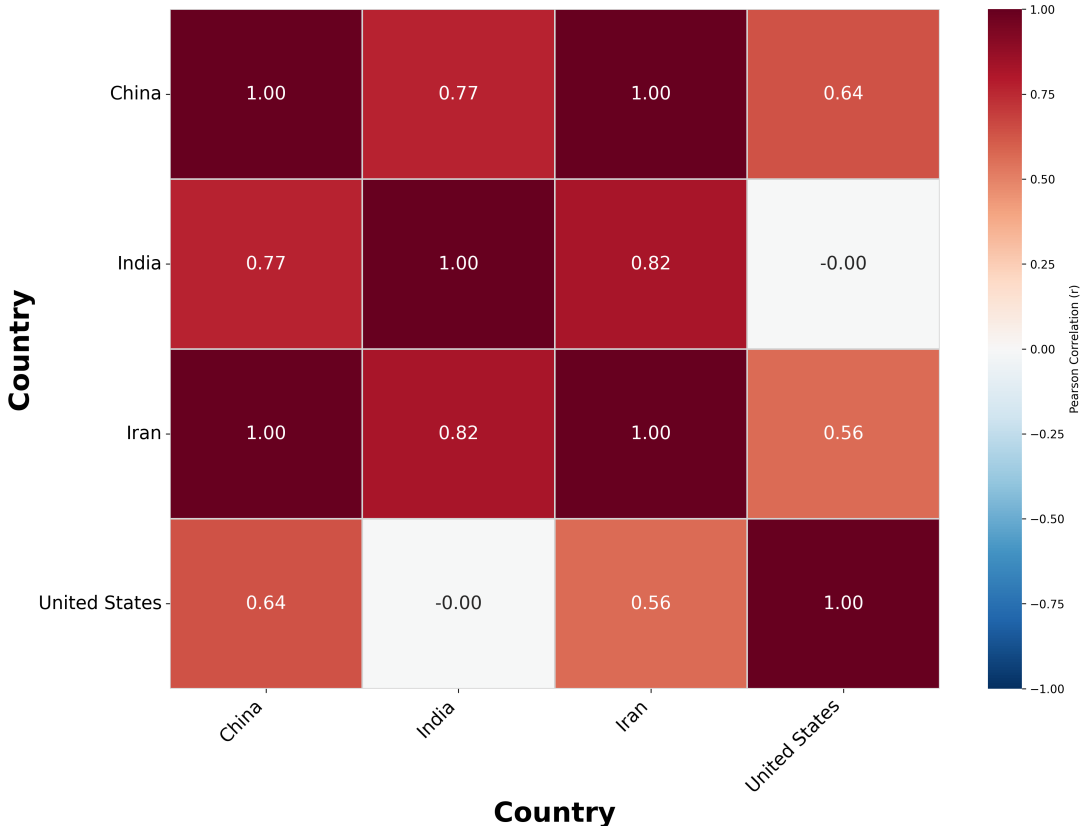


Figure 5: Country-level correlation matrix for Qwen 2.5-3B across BLEND, CulturalBench, and GeoMLAMA. Each cell shows the Pearson correlation between country pairs based on their accuracy scores across the three benchmarks (n=3 data points per pair). High correlations (e.g., China-Iran: $r = 1.00$) indicate countries show similar relative performance patterns across benchmarks.

profiles, Qwen shows zero correlation ($r = 0.00$), meaning India and US have completely independent trajectories across benchmarks for this model. For Llama, both countries show relatively consistent performance across benchmarks with similar patterns of relative strength and weakness. For Qwen, however, benchmarks that favor American cultural knowledge do not favor Indian cultural knowledge, suggesting fundamentally different evaluation approaches for these two cultural contexts.

This model-dependent variation in country-level patterns demonstrates that benchmark construct validity is not universal but emerges from the specific interaction between model training, cultural context, and benchmark design. The fact that China-Iran correlations remain consistently high across both models while India-US correlations vary dramatically (from 0.94 to 0.00) suggests that certain

cultural pairings may be more stable across models than others. However, we note an important confounding factor: culturally heterogeneous countries like India may be evaluated differently across benchmarks depending on which regional subcultures or linguistic communities are tested. If different benchmarks assess different Indian subcultures (e.g., Assamese versus Hindi-dominant culture), the "India" label conflates distinct cultural contexts, potentially contributing to the observed variability in country-level correlations. Disentangling this effect from genuine benchmark divergence would require explicit documentation of which specific subcultures each benchmark evaluates.

5.2.2 Performance Patterns Across BLEND, CulturalBench, and DLAMA-v1

Figures 7 and 8 present country correlation matrices for both models across BLEND, CulturalBench,

Country Correlation Matrix Model: Llama 3.2-3B (CulturalBench, BLEND, GeoMLAMA, 3 benchmarks)

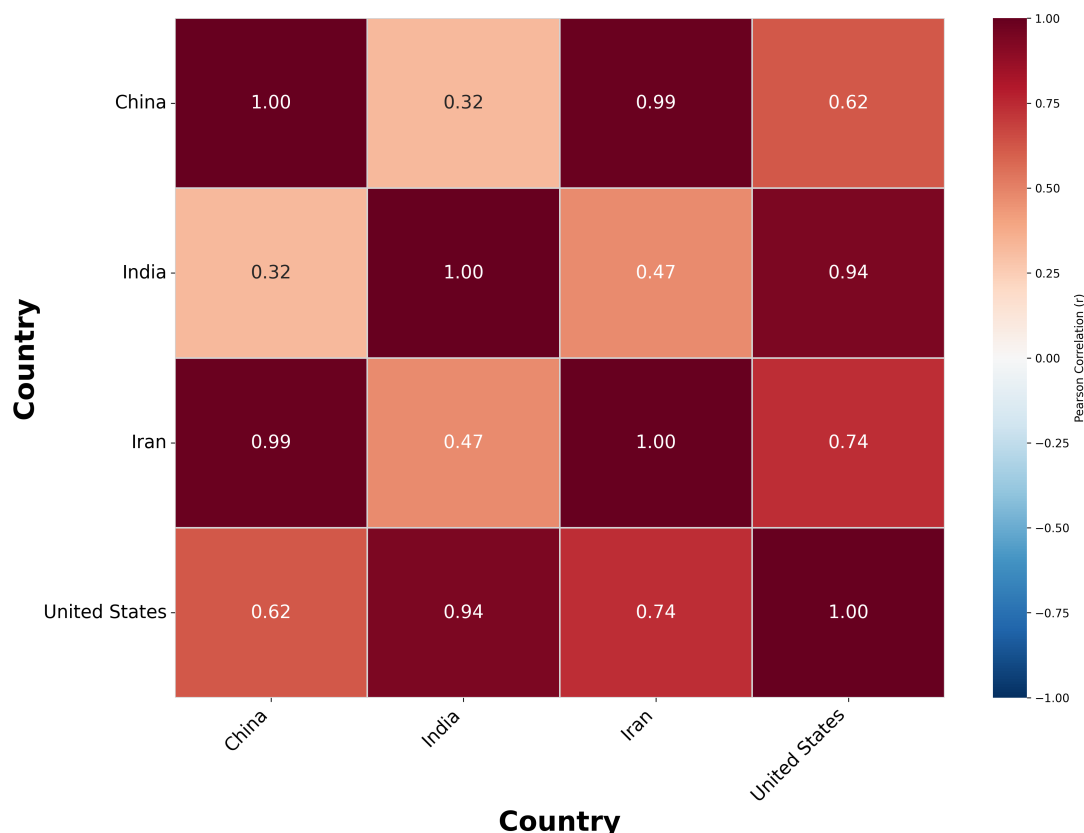


Figure 6: Country-level correlation matrix for Llama 3.2-3B across BLEND, CulturalBench, and GeoMLAMA. Note the strong China-Iran correlation ($r = 0.99$) and India-US correlation ($r = 0.94$), contrasting with weaker cross-regional correlations.

and DLAMA-v1. With six countries in this comparison (versus four for GeoMLAMA), the matrices reveal strikingly uniform patterns: nearly all country pairs exhibit very strong to perfect correlations ($r > 0.90$), with only minor variation between models.

Qwen 2.5-3B exhibits remarkably consistent patterns across all country pairs:

- **Spain-United Kingdom:** Perfect correlation ($r = 1.00$), indicating identical performance trajectories across all three benchmarks
- **Near-perfect correlations:** Most other country pairs show $r = 0.96$ - 0.99 , including China-Indonesia ($r = 0.99$), China-Spain ($r = 0.99$), and UK-US ($r = 0.98$)
- **Weakest correlation:** South Korea-United States ($r = 0.85$), still in the strong range

Llama 3.2-3B shows even more uniformity:

- **Perfect correlations:** Three country pairs reach $r = 1.00$ (China-South Korea, Spain-UK, UK-US), with two reaching statistical significance despite small sample size ($n=3$)
- **Near-perfect correlations:** All other pairs show $r = 0.94$ - 0.99
- **No weak correlations:** Even the lowest correlation (Indonesia-South Korea: $r = 0.91$) is substantially stronger than many correlations observed in the GeoMLAMA comparison

This pattern contrasts sharply with the GeoMLAMA comparison, where country-level correlations varied dramatically (e.g., India-US: $r = 0.94$ for Llama but $r = 0.00$ for Qwen). For the DLAMA comparison, nearly all countries show parallel performance patterns across the three benchmarks, regardless of model or geographic region. Western countries like the UK and US show strong

Country Correlation Matrix
Model: Qwen 2.5-3B
(CulturalBench, BLEND, DLAMA, 3 benchmarks)

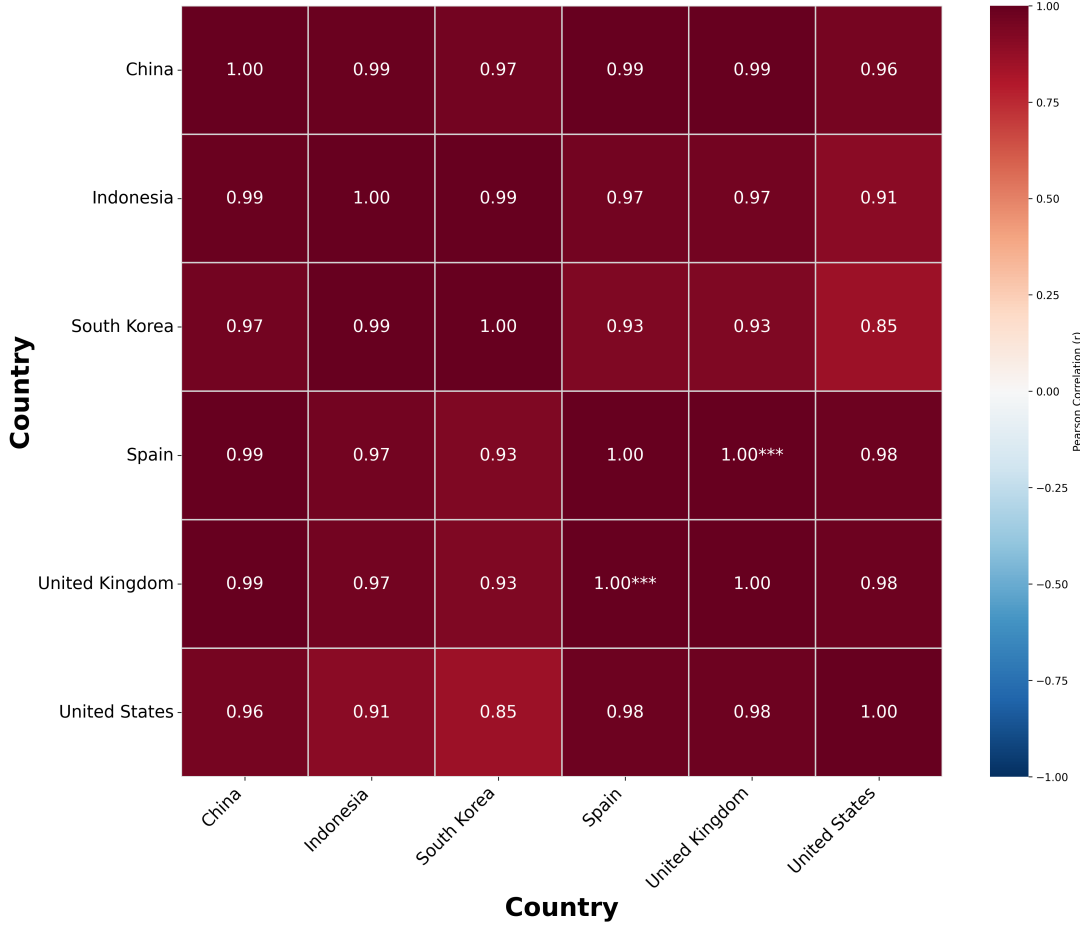


Figure 7: Country-level correlation matrix for Qwen 2.5-3B across BLEND, CulturalBench, and DLAMA-v1. Nearly all country pairs show very strong correlations ($r > 0.90$), with Spain-UK showing perfect correlation ($r = 1.00^{***}$).

correlations with Asian countries like China and South Korea ($r = 0.95$ - 1.00), suggesting that when DLAMA-v1 is included in the benchmark set, all evaluated countries experience similar relative ups and downs.

The uniformly high correlations likely reflect DLAMA-v1’s generation-based format creating a consistent difficulty layer across countries, resulting in more homogeneous performance patterns than observed with GeoMLAMA.

However, we note an important caveat: the six countries in this comparison (China, Indonesia, South Korea, Spain, UK, US) include more Western and East/Southeast Asian representation than the GeoMLAMA set, which included Iran. The absence of Middle Eastern representation and the predominance of economically developed nations

may contribute to the observed uniformity. Additionally, with only $n=3$ benchmarks per correlation, even small variations in one benchmark’s results would substantially impact correlation estimates, potentially inflating apparent similarity.

5.3 Discussion for Comprehensive Cultural Evaluation

Our findings have several important implications for cultural evaluation practices and future benchmark development.

5.3.1 Benchmarks Capture Distinct Facets, Not Unified Constructs

The weak-to-moderate correlations ($r < 0.5$ in many cases, with a minimum of $r = -0.261$) demonstrate that these benchmarks measure distinct aspects of cultural knowledge rather than a unified

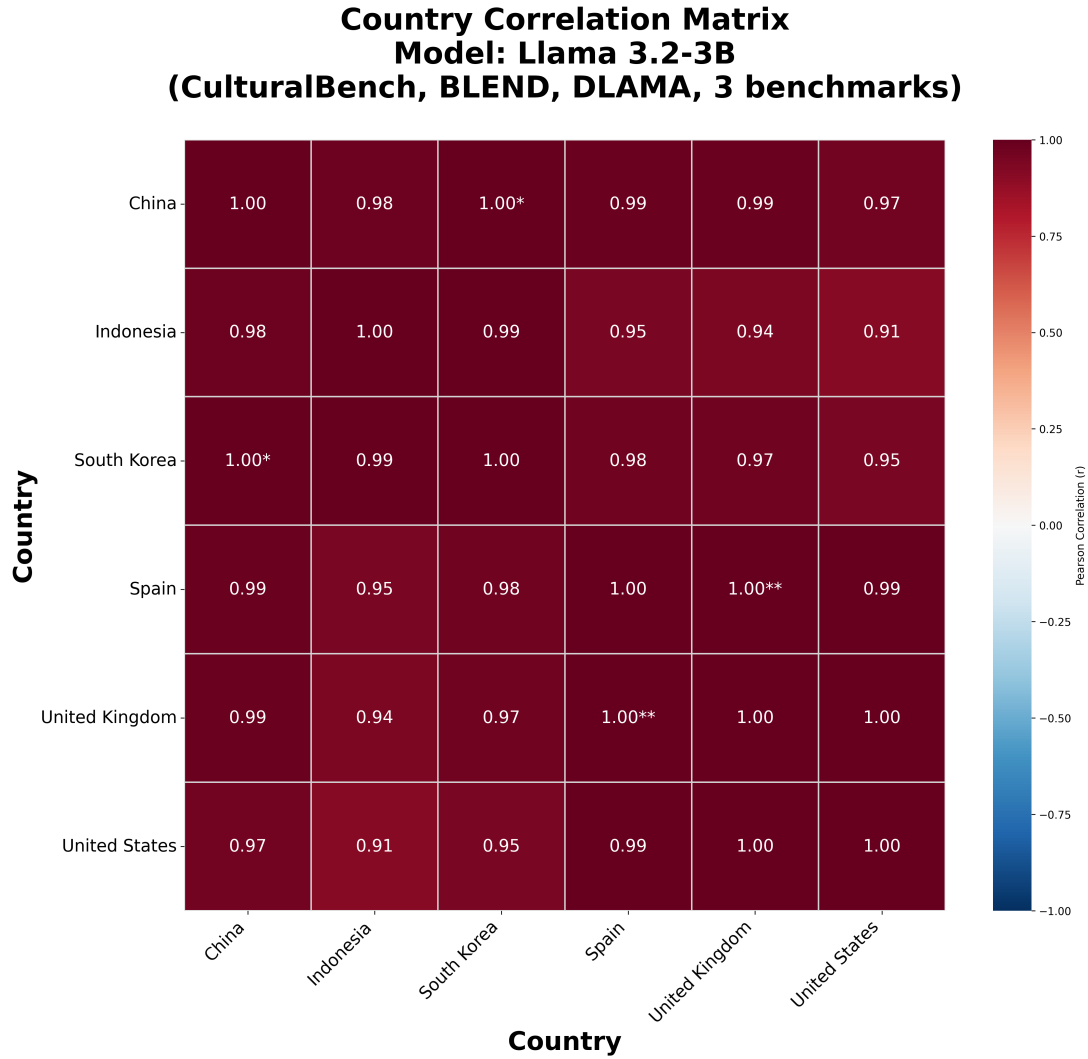


Figure 8: Country-level correlation matrix for Llama 3.2-3B across BLEND, CulturalBench, and DLAMA-v1. Similar to Qwen, nearly all country pairs show very strong correlations, with China-South Korea, Spain-UK, and UK-US all showing perfect correlations ($r = 1.00$).

construct, challenging assumptions of interchangeability. Models may excel in everyday practices (BLEND) while struggling with geographically-grounded facts (GeoMLAMA).

This finding aligns with theoretical work by AlKhamissi et al. (2024) arguing that cultural benchmarks operationalize culture through divergent lenses (culture-as-knowledge, culture-as-preference, culture-as-dynamics). Our empirical evidence demonstrates that even within the narrower category of culture-as-knowledge (common-sense cultural facts), benchmarks still measure non-interchangeable dimensions.

5.3.2 Model-Dependent Construct Validity

Benchmark relationships vary dramatically across models. The same benchmark pair shows correlations ranging from $r = -0.261$ (Llama) to $r = 0.507$

(Qwen) for BLEND-GeoMLAMA, a 0.77-point shift. This means benchmark validity depends on which model is evaluated, not just the benchmarks themselves. Researchers cannot assume benchmark relationships established with one model will hold for others, and validity claims should specify the model family tested.

5.3.3 Geographic Coverage Limitations

The necessity of restricting our analysis to only 4 countries (for GeoMLAMA comparison) or 6 countries (for DLAMA comparison) due to limited geographic overlap is itself a significant finding. This constraint highlights a critical gap in current benchmark design: even benchmarks that claim multicultural coverage often evaluate largely non-overlapping sets of regions, making systematic cross-benchmark validation difficult.

The limited overlap means our correlation estimates have wide confidence intervals and may not generalize to the 45+ regions covered across these benchmarks. Future benchmarks should include more overlapping countries to enable robust validity studies. Current gaps in geographic overlap exclude underrepresented regions from cross-benchmark analysis, further marginalizing them in evaluation practices.

5.3.4 Multi-Benchmark Evaluation Necessity

Given the weak correlations and model-dependent patterns, a comprehensive cultural assessment requires multiple complementary benchmarks. Single-benchmark evaluation risks missing cultural knowledge gaps, drawing narrow conclusions about "cultural knowledge," and overlooking model-specific biases (e.g., Llama's negative BLEND-GeoMLAMA correlation).

This aligns with [Shelby et al. \(2025\)](#)'s framework on user harms. Single-benchmark evaluation may miss quality-of-service, allocative, or representational harms, potentially allowing biased models to disproportionately harm certain user populations.

5.3.5 Culturally-Contingent Convergence

Benchmark convergence varies by cultural context and benchmark combination. In the GeoMLAMA comparison, some country pairs show near-perfect correlations (China-Iran: $r = 0.99-1.00$) while others show zero correlation (India-US for Qwen: $r = 0.00$). In contrast, the DLAMA comparison shows uniformly high correlations ($r > 0.90$) across all country pairs for both models. This indicates that benchmark validity is both region-specific and benchmark-set dependent. Researchers should report performance by cultural context rather than only aggregate metrics.

GeoMLAMA produces divergent country-pair correlations while DLAMA produces uniformly high correlations, indicating that benchmark composition affects cross-cultural comparisons. Future work should explore whether geographic proximity, linguistic similarity, or cultural characteristics predict these correlation patterns. Notably, the GeoMLAMA set includes Middle Eastern representation (Iran) while the DLAMA set does not (due to deadlines, we could not explore further this section of their benchmark), highlighting how regional diversity in evaluated countries may influence correlation structures.

6 Conclusion

We conducted the first systematic empirical investigation of whether cultural commonsense knowledge benchmarks measure the same underlying constructs. Through correlation analysis of four prominent benchmarks (BLEND, CulturalBench, GeoMLAMA, and DLAMA-v1) across two 3B-parameter multilingual models, we found weak to moderate benchmark-level correlations (Pearson $r < 0.5$ in most cases, with three negative correlations), demonstrating that these benchmarks capture distinct facets of cultural knowledge rather than a unified construct.

Our key findings reveal: (1) benchmark validity is model-dependent, the same benchmark pair shows correlations ranging from $r = -0.261$ to $r = 0.507$ across different models; (2) CulturalBench and BLEND strongly converge ($r = 0.895-0.899$), while DLAMA-v1 and GeoMLAMA diverge substantially from other benchmarks; (3) country-level correlations vary dramatically, with some pairs showing near-perfect correlations (China-Iran: $r = 1.00$) and others showing zero correlation (India-US for Qwen: $r = 0.00$); and (4) limited geographic overlap (only 4-6 common countries) constrains statistical power and excludes underrepresented regions.

These results challenge the assumption that cultural benchmarks are interchangeable. Comprehensive cultural assessment requires multiple benchmarks, as models may excel on everyday practices (BLEND) while struggling on geographically-grounded facts (GeoMLAMA). Without understanding which dimensions different benchmarks assess, we cannot identify when models systematically fail for certain user populations, missing quality-of-service, allocative, and representational harms.

Future work should develop benchmarks with broader overlapping country sets to enable robust cross-benchmark validation. Further qualitative analysis examining question types, knowledge domains, and evaluation formats could explain why correlations remain weak. As LLMs are deployed globally, rigorous methods for comprehensive cultural evaluation are essential for ensuring equitable service across diverse populations.

Limitations

This study has several important limitations that should be considered when interpreting our find-

ings.

Model Selection: We evaluate only two 3B-parameter models (Llama 3.2-3B and Qwen 2.5-3B) due to computational constraints. Larger models may exhibit different correlation patterns across benchmarks, and the limited model diversity ($n=2$) restricts the statistical power of any aggregate benchmark-level comparisons. Future work should examine whether our findings generalize to larger model families and architectures.

Benchmark and Geographic Coverage: Our analysis examines four benchmarks within the common-sense knowledge category, excluding frameworks assessing cultural values, norms, or pragmatic reasoning. Additionally, sparse geographic overlap across benchmarks (4-6 common countries representing a small fraction of 45+ regions covered) limits statistical power and prevents comprehensive four-way comparisons. This constraint may not capture correlation patterns that would emerge with broader coverage.

Evaluation Methodology: We evaluate most benchmarks in English to isolate cultural knowledge from language proficiency, but this approach does not reflect authentic multilingual cultural contexts. Additionally, our country-level correlation analysis treats each country as an independent data point, which may not account for shared cultural characteristics among neighboring or historically connected regions. The minimum threshold of 10 samples per country may exclude some regions with valuable but limited representation.

Qualitative Analysis: We do not investigate the specific semantic or structural features that drive benchmark differences, such as question phrasing, answer format, or the types of cultural knowledge being tested (factual vs. experiential vs. normative). Such qualitative analysis would provide deeper insights into why correlations are weak but requires extensive manual annotation beyond the scope of this work and semester deadlines.

Binary Accuracy Metric: Our use of binary accuracy (correct/incorrect) does not capture partial correctness, cultural nuance, or the severity of errors. A model answering "rice" instead of "biryani" for an Indian dish question represents a different type of error than answering "pizza," but both receive the same score of 0. Future works should

further record this.

Ethical Considerations

Cultural knowledge evaluation carries important ethical considerations that we address throughout this work.

Representation and Stereotyping: Benchmarks may inadvertently reinforce stereotypes or oversimplify complex cultural practices. Cultural knowledge is dynamic, context-dependent, and often contested within communities. Our analysis examines benchmark construct validity rather than defining correct cultural knowledge. Researchers should recognize these limitations and avoid treating benchmark performance as a definitive measure of cultural competence.

Geographic Coverage Limitations: Our correlation analysis reveals a critical limitation: only 4-6 countries have overlapping coverage across benchmark pairs, forcing us to exclude the vast majority of evaluated regions from cross-benchmark comparison. This methodological constraint, while necessary for rigorous correlation analysis, risks further marginalizing already underrepresented regions in evaluation practices. The geographic coverage gaps in existing benchmarks highlight the urgent need for more comprehensive multicultural evaluation resources with broader regional overlap.

Evaluation Language and Access: We evaluate all benchmarks in English, which may disadvantage models' access to cultural knowledge that is more naturally expressed in local languages. While this choice isolates cultural knowledge from language proficiency for comparison purposes, it does not reflect how cultural knowledge is actually acquired, stored, or expressed in multilingual contexts. Future work should examine how language choice affects cultural knowledge assessment.

LLM-as-Judge Limitations: Our use of an LLM judge (OLMo-2-1124-7B-Instruct) for DLAMA-v1 evaluation introduces potential biases, as judge models may have their own cultural knowledge gaps or preferences. While we designed our judge prompts to handle dialectal variations and synonyms, we acknowledge that judge model hallucinations and errors cannot be completely avoided, potentially introducing noise into evaluation results.

Western-Centric Training Data: The models we evaluate (Llama 3.2-3B and Qwen 2.5-3B) are trained on predominantly Western-centric data. Our findings of weak cross-benchmark correlations and performance variations across cultural contexts reflect not inherent properties of cultures, but rather systematic biases in model training data and benchmark design choices.

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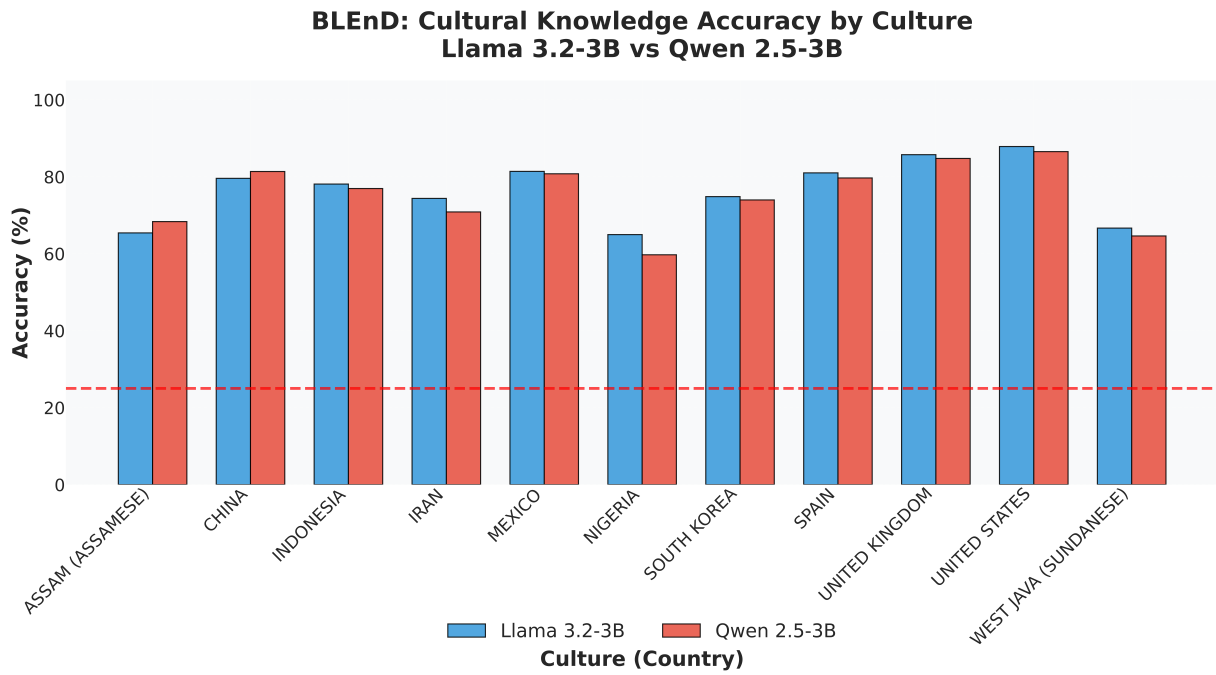


Figure 9: Accuracy by country for BLEnD benchmark across both evaluated models.

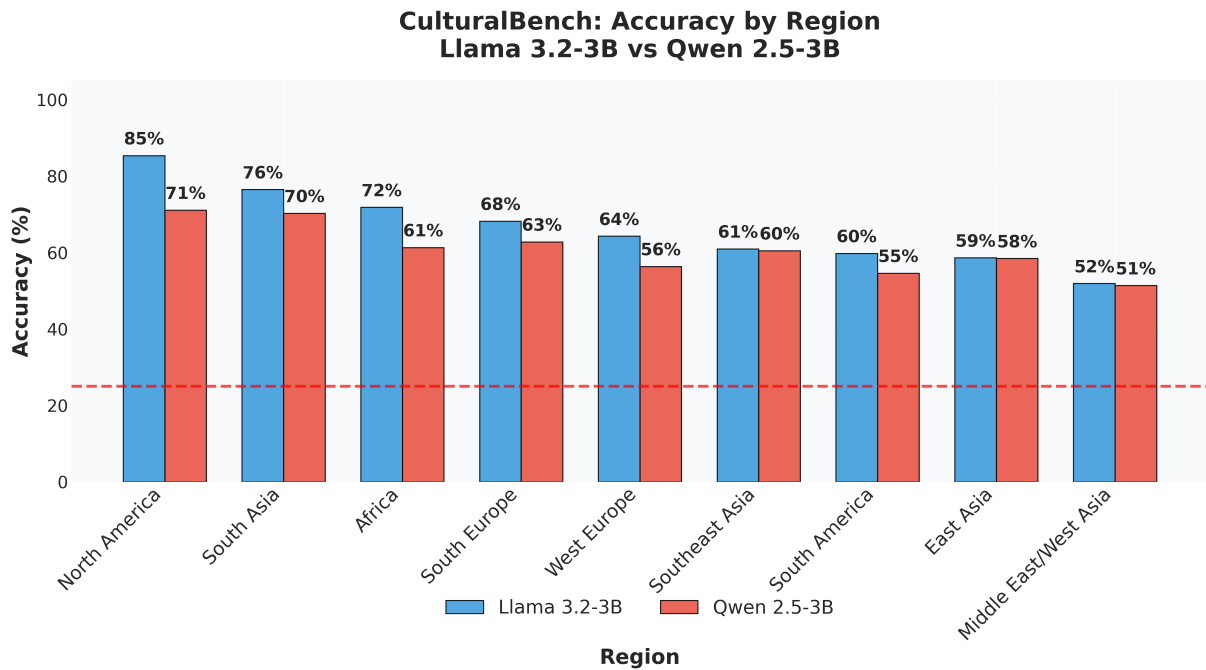


Figure 10: Accuracy by region for CulturalBench benchmark across both evaluated models.

Table 1: Accuracy by region for CulturalBench benchmark.

Region	Llama 3.2-3B	Qwen 2.5-3B
North America	85.36%	71.07%
South Asia	76.46%	70.25%
Africa	71.81%	61.30%
South Europe	68.19%	62.78%
West Europe	64.26%	56.35%
Southeast Asia	60.94%	60.47%
South America	59.75%	54.58%
East Asia	58.60%	58.44%
Middle East/West Asia	51.90%	51.37%
AVERAGE	66.36%	60.74%

Table 2: Accuracy by country for CulturalBench benchmark.

Region	Country	Llama 3.2-3B	Qwen 2.5-3B
Africa	Zimbabwe	88.24%	76.47%
	Nigeria	81.82%	68.18%
	South Africa	79.31%	70.69%
	Morocco	64.71%	41.18%
	Egypt	45.00%	50.00%
East Asia	Japan	71.70%	69.81%
	Hong Kong	66.67%	63.89%
	China	57.63%	57.63%
	South Korea	56.10%	46.34%
	Taiwan	40.91%	54.55%
Middle East/W. Asia	Turkey	65.79%	63.16%
	Lebanon	54.55%	50.00%
	Saudi Arabia	47.06%	41.18%
	Israel	46.15%	53.85%
	Iran	45.95%	48.65%
North America	Canada	85.71%	57.14%
	United States	85.00%	85.00%
South America	Mexico	67.35%	63.27%
	Peru	63.16%	57.89%
	Chile	59.09%	45.45%
	Argentina	57.14%	54.29%
	Brazil	52.00%	52.00%
South Asia	Nepal	85.71%	80.95%
	Pakistan	78.57%	85.71%
	Bangladesh	72.00%	60.00%
	India	69.57%	54.35%
South Europe	Spain	72.50%	70.00%
	Italy	63.89%	55.56%
Southeast Asia	Thailand	77.78%	74.07%
	Malaysia	63.64%	54.55%
	Singapore	60.87%	56.52%
	Philippines	57.78%	53.33%
	Vietnam	55.56%	66.67%
	Indonesia	50.00%	57.69%
West Europe	United Kingdom	76.00%	72.00%
	Germany	68.75%	56.25%
	France	64.29%	57.14%
	Netherlands	48.00%	40.00%

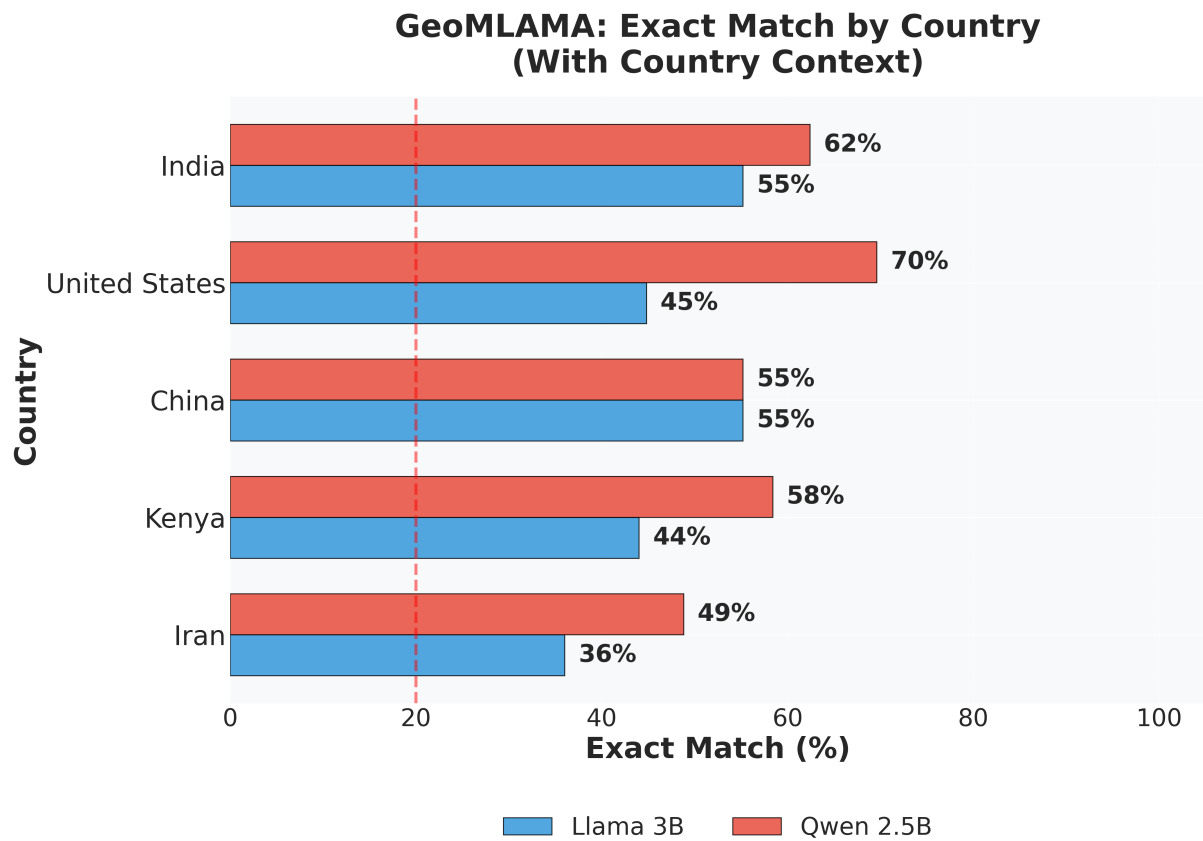


Figure 11: Country-level accuracy for GeoMLAMA benchmark across both evaluated models.

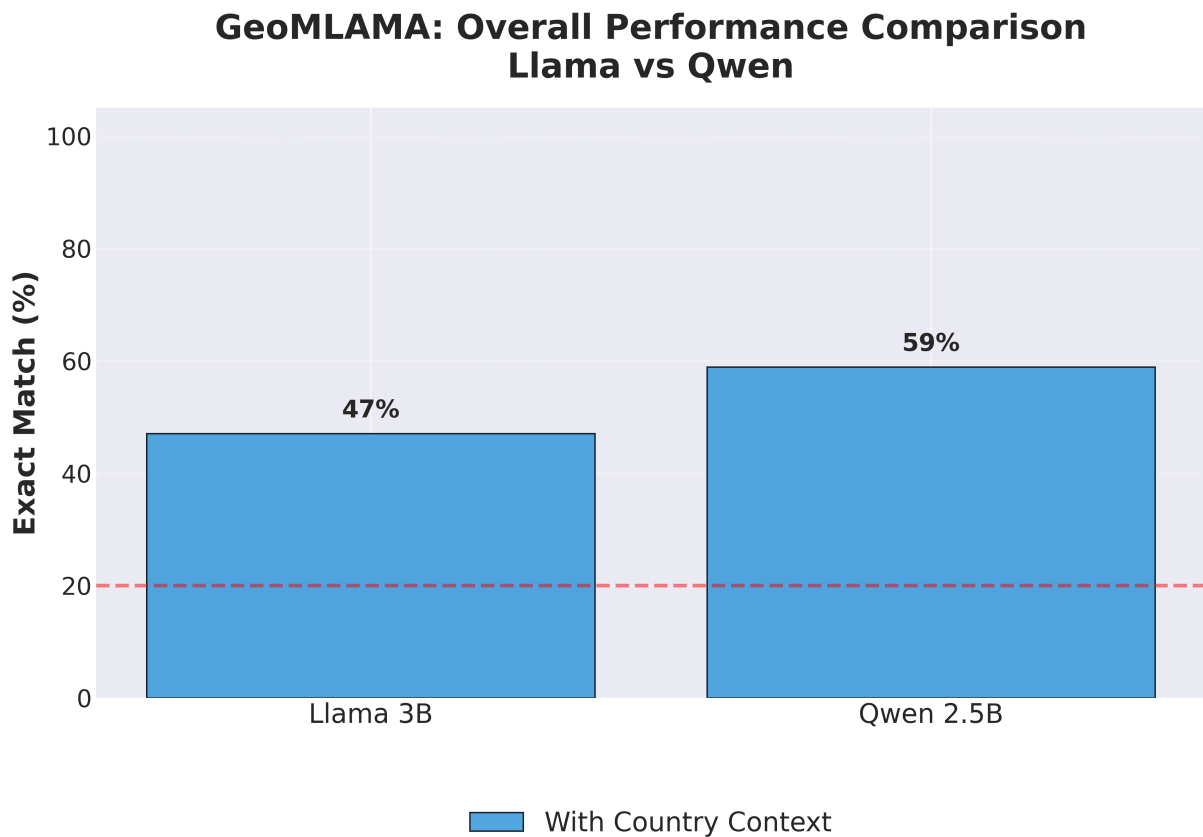


Figure 12: Overall accuracy comparison for GeoMLAMA benchmark.

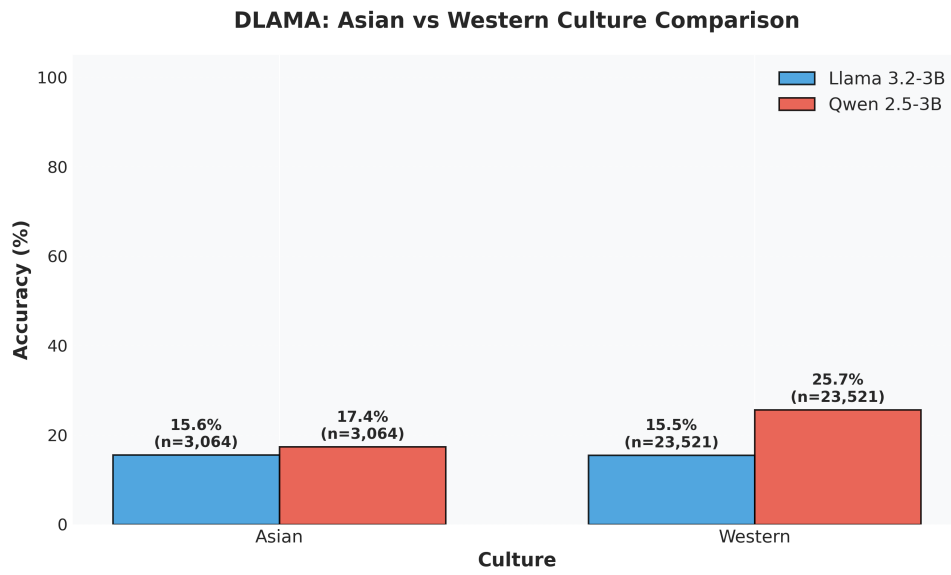


Figure 13: DLAMA-v1 regional comparison between Asian and Western cultures.

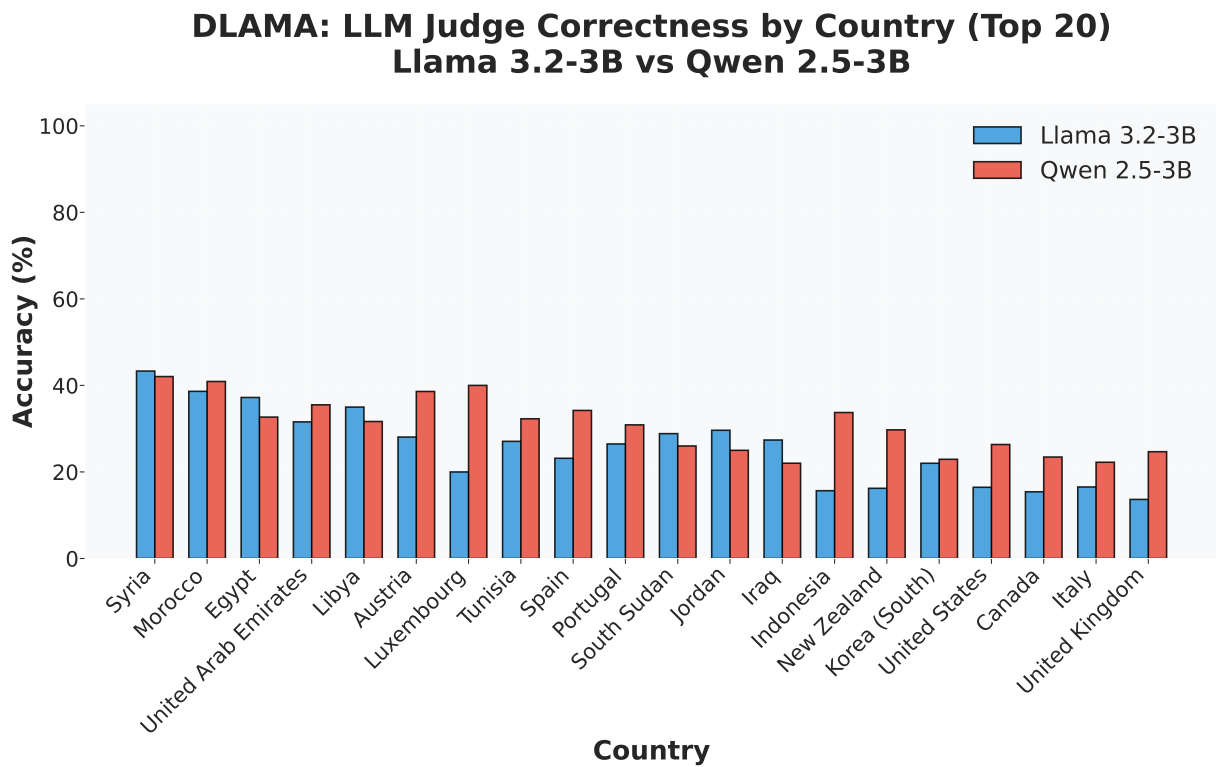


Figure 14: Country-level accuracy for DLAMA-v1 benchmark across both evaluated models.

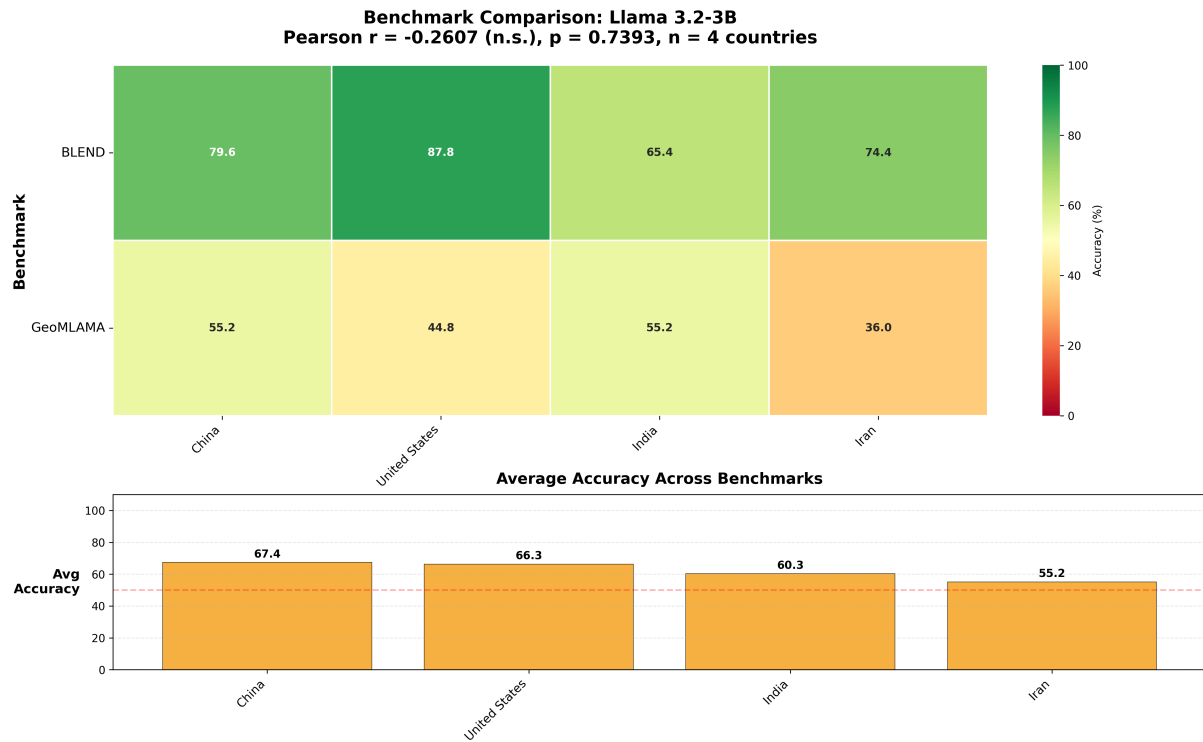


Figure 15: Benchmark comparison between BLEND and GeoMLAMA for Llama 3.2-3B across 4 common countries. Top panel shows accuracy heatmap; bottom panel shows average performance. The negative correlation ($r = -0.261$) is visible in the inverse ranking patterns: United States ranks 1st on BLEND but 2nd on GeoMLAMA, while China ranks 3rd on BLEND but 1st on GeoMLAMA.

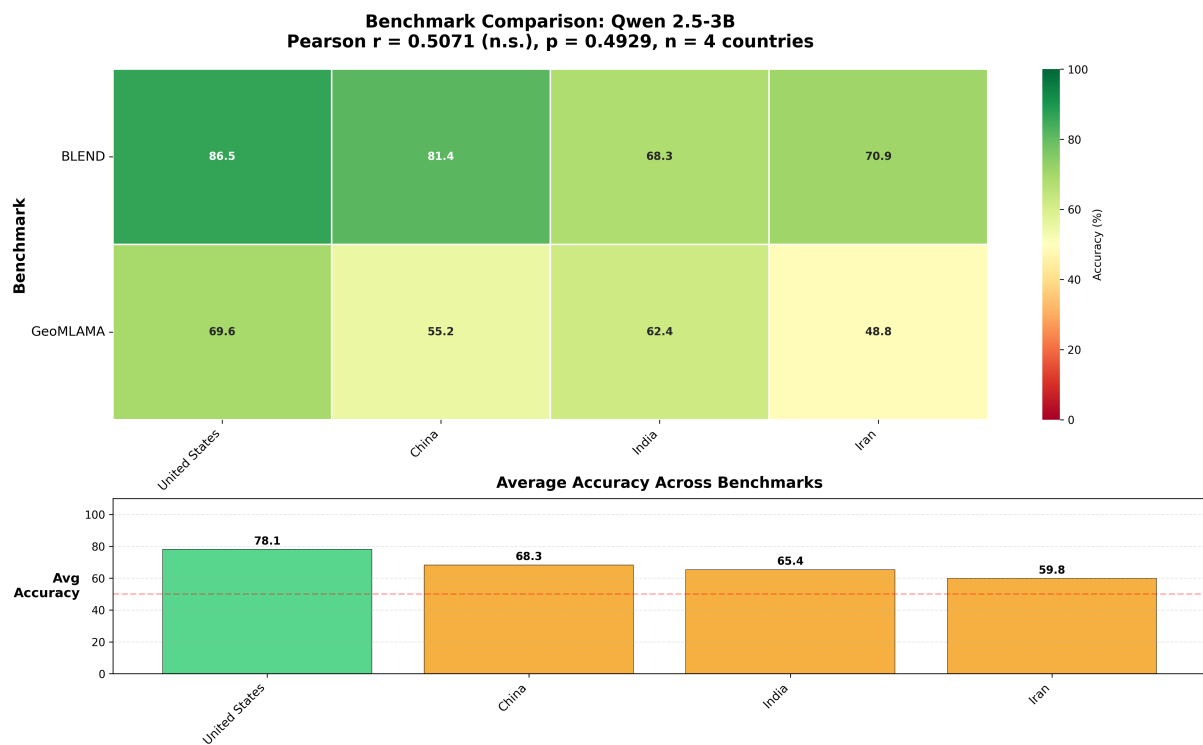


Figure 16: Benchmark comparison between BLEND and GeoMLAMA for Qwen 2.5-3B across 4 common countries. Unlike Llama's negative correlation, Qwen shows moderate positive correlation ($r = 0.507$), with more consistent country rankings across both benchmarks.